



PRS Pricing Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 23-March-2026

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1 Test Results — PRS Pricing Model (MKM-PR-001)

Tests run on **23 March 2026 at 10:09:55**.

Table 1: Test Results Summary — PRS Pricing Model (MKM-PR-001)

Total Tests	61
Passed	61
Failed	0
Skipped	0
Duration	10.98s

Table 2: Detailed Test Results — PRS Pricing Model

Test Description	Acceptance Criteria	Result	Time
Basis waterfall in generated output	<code>bw1["model.uncertainty_bp"] == 2.0;</code> <code>bw1["terrain_bp"] == 2.0 (+6 more)</code>	Pass	0.875s
Composition flat penalty	<code>result["composition_bp"] ==</code> <code>COMPOSITION_BASIS_BPS["Flat"]</code>	Pass	0.000s
Composition pre2000 penalty	<code>result["composition_bp"] == 1.0</code>	Pass	0.000s
Distance basis capped	<code>result["distance_bp"] ==</code> <code>DISTANCE_MAX_BPS</code>	Pass	0.000s
Distance basis scales linearly	<code>result["distance_bp"] ==</code> <code>pytest.approx(1.0, abs=0.1)</code>	Pass	0.000s
Elevation above gauge reduces spread	<code>result["elevation_bp"] ==</code> <code>pytest.approx(-1.0, abs=0.1)</code>	Pass	0.000s
Elevation below gauge no benefit	<code>result["elevation_bp"] == 0.0</code>	Pass	0.000s
Elevation capped at 3bp	<code>result["elevation_bp"] ==</code> <code>-ELEVATION_MAX_BENEFIT_BPS</code>	Pass	0.000s
Model uncertainty is fixed	<code>result["model.uncertainty_bp"] ==</code> <code>MODEL_UNCERTAINTY_BPS</code>	Pass	0.000s
Terrain zone1 low risk	<code>result["terrain_bp"] == 0.0</code>	Pass	0.000s
Terrain zone3 high risk	<code>result["terrain_bp"] ==</code> <code>TERRAIN_BASIS_BPS["Zone 3a"]</code>	Pass	0.000s
Total basis sums components	<code>result["total.basis_bp"] ==</code> <code>pytest.approx(expected_total, ...)</code>	Pass	0.000s
1Y from Feb 2026 = last Friday May 2027.	<code>mat == last_friday_of_month(2027, 5)</code>	Pass	0.000s
3Y from Feb 2026 = last Friday May 2029.	<code>mat == last_friday_of_month(2029, 5);</code> <code>mat.month == 5 (+1 more)</code>	Pass	0.000s
Jan 2027 uses Aug 2026 roll → Nov 2029 maturity.	<code>mat == last_friday_of_month(2029, 11)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).	<code>mat == last_friday_of_month(2029, 11);</code> <code>mat.month == 11</code>	Pass	0.000s
5Y from Feb 2026 = last Friday May 2031.	<code>mat == last_friday_of_month(2031, 5)</code>	Pass	0.000s
5Y from Sep 2026 = last Friday Nov 2031.	<code>mat == last_friday_of_month(2031, 11)</code>	Pass	0.000s
All maturities should fall on a Friday.	<code>mat.weekday() == 4</code>	Pass	0.000s
3Y maturity jumps from May to Nov at the Aug roll.	<code>may_mat.month == 5; nov_mat.month == 11 (+1 more)</code>	Pass	0.000s
Aug onwards uses Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8</code>	Pass	0.000s
December uses the same year's Aug roll.	<code>roll.month == 8; roll.year == 2026</code>	Pass	0.000s
Feb-Jul uses Feb roll.	<code>roll == last_friday_of_month(2026, 2);</code> <code>roll.month == 2</code>	Pass	0.000s
January uses the previous year's Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8 (+1 more)</code>	Pass	0.000s
July is the last month of the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s
March is still in the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s
<code>compute_maturity_date</code> with no <code>ref_date</code> uses today.	<code>isinstance(mat, date); mat.weekday() == 4</code>	Pass	0.000s
<code>current_roll_date(None)</code> uses today and returns a valid date.	<code>isinstance(roll, date);</code> <code>roll.weekday() == 4</code>	Pass	0.000s
<code>maturity_table</code> with no <code>ref_date</code> uses today.	<code>len(table) == 5;</code> <code>all(isinstance(e['maturity_date'], date) for e in table)</code>	Pass	0.000s
Empty propertyts dir	<code>stats["total_properties"] == 0;</code> <code>stats["properties_with_gev"] == 0</code>	Pass	0.002s
Generator should still work without gauge hazard curves (empty basis).	<code>stats["properties_with_gev"] == 2;</code> <code>stats["properties_with_floor"] == 1</code> <code>(+1 more)</code>	Pass	0.872s
No propertyts dir	—	Pass	0.000s
Aug 2026: 31st is Monday, last Friday = 28th.	<code>last_friday_of_month(2026, 8) == date(2026, 8, 28)</code>	Pass	0.000s
Feb 2026: 28th is Saturday, last Friday = 27th.	<code>last_friday_of_month(2026, 2) == date(2026, 2, 27)</code>	Pass	0.000s
May 2029: 31st is Thursday, last Friday = 25th.	<code>last_friday_of_month(2029, 5) == date(2029, 5, 25)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Nov 2029: 30th is Friday.	<code>last_friday_of_month(2029, 11) == date(2029, 11, 30)</code>	Pass	0.000s
Result should always be a Friday (weekday=4).	<code>d.weekday() == 4</code>	Pass	0.000s
Actual years should be strictly increasing.	<code>years[i] > years[i - 1]</code>	Pass	0.000s
All maturities should be in May for the Feb roll.	<code>e['maturity_date'].month == 5</code>	Pass	0.000s
All maturities should be in November for the Aug roll.	<code>e['maturity_date'].month == 11</code>	Pass	0.000s
Maturity string should be formatted as 'DD Mon YYYY'.	<code>len(e['maturity_str']) > 8; 'May' in e['maturity_str'] or 'Nov' in e['maturity_str']</code>	Pass	0.000s
Default table has 5 entries (1Y through 5Y).	<code>len(table) == 5</code>	Pass	0.000s
Tenors should be 1, 2, 3, 4, 5.	<code>tenors == [1, 2, 3, 4, 5]</code>	Pass	0.000s
Positive spread	<code>spread > 0; 50 < spread < 500</code>	Pass	0.000s
Spread increases with hazard	<code>spread_high > spread_low</code>	Pass	0.000s
Spread values at different tenors	<code>s > 0</code>	Pass	0.000s
Zero hazard rate	<code>spread == 0.0</code>	Pass	0.000s
Basis calculation	<code>len(nearest) > 0; ng0["gauge_id"] == "GAUGE-001" (+5 more)</code>	Pass	0.963s
Depth thresholds	<code>"any_flood" in thresholds; "moderate" in thresholds (+4 more)</code>	Pass	0.872s
Generate produces output	<code>output_path.exists(); "metadata" in data (+2 more)</code>	Pass	0.859s
Gev params structure	<code>"shape" in gev; "loc" in gev (+2 more)</code>	Pass	0.863s
High transmission rate	<code>prop3["summary"]["flood.transmission_rate"] == 1.0</code>	Pass	0.981s
Output structure	<code>"PROP-001" in curves; "PROP-002" in curves (+21 more)</code>	Pass	0.867s
Properties with gev	<code>stats["properties_with_gev"] == 2; stats["properties_with_floor"] == 1 (+1 more)</code>	Pass	0.871s
Survival decreases with tenor	<code>survival[i] <= survival[i - 1]</code>	Pass	0.875s
Term structure	<code>ts["tenors"] == TENORS; threshold_name in ts (+3 more)</code>	Pass	1.009s
Recovery rates by trigger	<code>len(spreads) > 0</code>	Pass	1.067s
Recovery reduces spread	<code>spread_with_recovery < spread_no_recovery</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All tenors should round-trip through maturity and back.	<code>tenor_from_maturity(mat, ref) == t</code>	Pass	0.000s
A date not in the table returns the closest tenor.	<code>result in [1, 2, 3, 4, 5]</code>	Pass	0.000s
Exact maturity date returns correct tenor.	<code>tenor_from_maturity(mat, date(2026, 2, 24)) == 3</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (30 tests)	D. Kelly
07-Mar-2026	1.1	57 test(s) added; 30 test(s) removed (total: 57)	D. Kelly
07-Mar-2026	1.2	30 test(s) added (total: 87)	D. Kelly
18-Mar-2026	1.3	4 test(s) added (total: 91)	D. Kelly
23-Mar-2026	1.4	30 test(s) removed (total: 61)	D. Kelly